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PI Name:	Barcellos-Hoff, Mary Helen Ph.D.		
Project Title:	NSCOR: The contribution of non-targeted effects in HZE cancer risk		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Human Research Program Elements:	(1) **SR**:Space Radiation		
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Comments:	NOTE: Former affiliation was Lawrence Berkeley National Laboratory; moved to NYU in August 2008. Moved to UCSF September, 2015		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSCOR Space Radiation NNJ08ZSA003N
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
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Flight Program:			
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Task Description:	Clear mechanistic understanding of the biological processes elicited by high LET radiation increase cancer risk will reduce the uncertainties in assessing the risk of space flight. Classic targeted radiation effects like cell kill and mutation are proportional to dose and their LET dependence is described by relative biological effectiveness. However recent data suggests that high LET radiation affects multicellular processes that promote cancer. We propose that radiation carcinogenesis results from two classes of ionizing radiation effect: targeted effects that alter genomic sequence and non-targeted effects (NTE) that alter phenotype and multicellular interactions. Since NTE are not proportional to dose, new approaches are required to assess the impact of NTE, which alter gene expression, cell signaling, genomic stability and cell phenotypes, on carcinogenesis. It is unknown to what extent NTE contribute to the significantly greater carcinogenic effect of high LET radiation. We hypothesize that NTE promotes epithelial cancer. Project 1 will test use novel transgenic and humanized mouse mammary models to evaluate the relative contribution of radiation targeted and NTE to carcinogenesis as a function of radiation quality and of age a radiation chimera model. We will monitor inflammation as a key process affected by radiation NTE and conduct a systems biology analysis of mammary gland. Project 2 will employ new systems genetic analysis of cancer susceptibility in radiation sensitive organs to define critical regulatory networks that impact the relative carcinogenic effectiveness of high LET radiation. Project 3 will use Monte Carlo simulations of targeted radiation effects and agent based modeling of cell interactions to model the relative contributions of targeted and NTE to carcinogenesis. This NSCOR will provide important data on radiation quality dependence of targeted and non-targeted radiation effects and the genetic determinants of cancer susceptibility, which together will used to reduce uncertainties in NASA radiation risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	First, radiation can be considered an experimental probe that perturbs tissue function. Studying these perturbations is a means to define critical biological response to stress. Second, we will develop a comprehensive data base of normal tissue and irradiated tissue by analyzing different organs in each mouse. This data base will become a public resource. Third, our studies will identify genetic predisposition to radiation associated cancer, and may thus help protect humans who have the greatest risk of cancer.
	Our NSCOR provides experimental and modeling studies to define the efficiency and physiological context in which high LET radiation increases epithelial cancer risk. We hypothesize that both targeted and NTE contribute to radiation carcinogenesis. Targeted effects of ionizing radiation result from interaction of energy with DNA, leading to DNA damage responses and changes in genomic sequence. On the other hand, poorly understood NTE alter phenotype and multicellular interactions that could contribute to cancer. We hypothesize that the greater carcinogenic risk from high LET irradiation is due to NTE promotion of targeted radiation effects. The data generated the NYU/LBNL/UCSF NSCOR will provide a comprehensive analysis of LET dependence for solid tumor frequency in different tissues, integrated with computational modeling and systems biology to identify key events. The goal of Project 1 is to determine the contribution of NTE as a function of radiation quality on breast cancer. Aim 1.1 uses a mammary chimera model in which wildtype Balb/c mice are transplanted orthotopically with syngeneic Trp53 null mammary epithelium. Tumorigenesis was compared in mice irradiated with different particles before transplantation (i.e. the radiation chimera), in which epithelium is not exposed), irradiation of intact mammary glands (i.e. the genetic chimera, in which a Trp53 null epithelial outgrowth is irradiated in situ), and in aged mice in which 10 month old mice were irradiated before transplantation. Aim 1.2 investigates the inflammatory signals induced by radiation. Aim 1.3 uses systems biology to integrate data from different models and types to provide a better mechanistic understanding of underlying biology. NOTE: The Barcellos-Hoff and Demaria labs at NYU were heavily damaged by Hurricane Sandy on October 29th, 2012. As a consequence, all work ceased for 3 full months while we were relocated to a new laboratory in January. Several reagents were compromised by loss of power, in particular the Trp53 null tissue fragments needed for the transplantation experiments. As a consequence we are unable to conduct the low dose rate experiments planned for this spring and summer. Our current analyses show that NTE do contribute to carcinogenesis by affecting tumor latency, growth rate, and characteristics. Tumors affected only by radiation NTE, i.e. arising in irradiated mice, were far more aggressive than those from non-irradiated mice. In contrast tumors arising from irradiated Trp53 epithelium were distinct in that neither ER-negative tumors nor metastases were increases. We conclude that aggressive tumors are promoted by radiation NTE host responses such as oscillatory inflammation. We have not observed a robust LET dependence but given our evolving understanding of the NTE dose-dependence, it may well be that thresholds rather than proportional responses may be more critical. For example, it may be that each NTE has a distinct threshold and that these change as a function of LET. The effect on tumor type (e.g. ER statues) appears to be the most sensitive index in the radiation chimera model. This threshold affecting tumor type was exceeded with doses of 10 cGy (Nguyen et al., 2011) and with a fluence of 1 particle/100 microm2 of high LET particles. The lack of classical dose dependence is challenge in the design of animal experiments. Gene expression profiles of tumors that arose in irradiated mice are distinct from those that arose in naïve hosts. We asked whether expression metaprofiles could discern radiation-preceded human cancer or be informative in sporadic breast cancers. Human orthologs of the host irradiation metaprofile discriminated between radiation-preceded and sporadic human thyroid cancers. Detection of radiation-preceded human cancer by the irradiated host metaprofile raises possibilities of assessing human cancer etiology. This paper was published in Clinical Cancer Research in February as was accompanied by a perspective written by Dr. Barcellos-Hoff on the role of tissue microenvironment in determining aggressive tumor types. Project 2 has the ultimate goal of identifying the genes and pathways associated with susceptibility to cancer risk from high LET radiation. The strategies employed are based on previous experience in using systems genetics approaches to identify germ line polymorphisms and somatic genetic changes associated with tumor development induced by chemicals or ionizing (low LET) radiation. This project is therefore focused heavily on genetic analysis of radiation-induced tumors, and as such complements the goals of Project 1 on analysis of the effects of radiation exposure on the normal or tumor microenvironment. The tumor suppressor Trp53, which is regularly mutated in cancers, is often referred to as the guardian of the genome due to the multiplicity of anti-proliferative effects it can initiate to maintain genomic integrity under genotoxic stress

Task Progress:	conditions. Germline deletion of a small N-terminal proline-rich region of Trp53 in an in vivo mouse model (Trp53 delta P) causes no spontaneous tumors; yet, upon exposure of the Trp53 delta P mouse to both low and high LET radiation, a wide range of solid tissue tumors develops, including those of the liver, lung and kidney. This range is in stark contrast to the tumor spectrum in the Trp53 null mice post-radiation, which mainly comprise lymphomas and sarcomas. The Trp53 deltaP mouse therefore represents an exquisitely radiation-sensitive model to study radiation-induced tumors similar to those seen in the human population. We have previously identified a unique pattern of genomic alterations arising in low LET radiation-induced lymphomas in Trp53 null mice. In order to enable direct comparisons with our previous data we have successfully bred the Trp53 deltaP mice onto a 99.9% pure FVB/N background. We are investigating impact of high LET compared to low LET upon tumor latency and spectrum in mouse models with compromised Trp53 activity. We exposed 282 FVB/N mice of genotype homozygous Trp53 ?P, WT and Trp53 heterozygotes, to 50cGy of either 600 MeV/amu Fe particles or gamma radiation. Our results show a significant reduction in tumor latency in Trp53 delta P and Trp53 heterozygote mice exposed to high LET radiation, in comparison to mice of the same genotype exposed to low LET. Of note, there is a negligible impact of genotype (either Trp53 delta P or Trp53 heterozygotes) in tumor-free survival when mice are exposed to the same radiation quality. Unexpectedly, we have observed a high number of mammary tumors and skin carcinomas in the Trp53 heterozygote mice following exposure to high LET radiation. We have previously exposed Trp53 heterozygote mice on a range of different genetic backgrounds to various doses of gamma radiation and have not observed a high frequency of mammary or skin tumors. This suggests a radiation-quality specific shift in tumor spectrum. Further investigation of the potential mechanism by which exposure to high LET radiation causes this shift in spectrum and reduced latency is currently underway. In order to gain a greater insight into the acute cellular response mechanism to exposure to high LET radiation we have performed a series of short time point experiments, with time points ranging between 6 hours and 1 week post 600MeV/amu Fe particle radiation. Preliminary analysis of microarray data from RNA of the dorsal skin suggests heightened expression of gene clusters enriched for acute inflammatory response at the 6 hour time point in the Trp53 delta P mice compared to the WT, this difference is not seen in the later time points. Conversely, genes associated with proliferation have much lower expression in the Trp53 delta P compared to WT at this 6 hour time point. To investigate the possibility that the patterns of genetic changes in tumors induced by high and low LET radiation are different in Trp53 abrogated mice, we have begun an analysis of whole exome sequences of a range of tumors by radiation exposure. We have extracted DNA from 61 radiation-induced (either high or low LET) tumors of the mammary and skin, and thymic lymphomas in Trp53 delta P and Trp53+/- mice and sent these, along with normal control DNA samples, to our collaborator David Adams at the Sanger Institute, Cambridge, UK to carry out exome sequencing. Our data have identified candidate mutations in genes such as Notch1, Pten, Hras, Kras and Trp53, but further validation and confirmation of these mutations, as well as detailed analysis of the mutation spectrum, is in progress. Project 3 uses agent-based models (ABM) to simulate the dynamic response of tissue to radiation in conjunction with Monte Carlo models of DNA damage to provide an integrated mechanistic model of the biology. This modeling approach is used to evaluate and prioritize key events during radiation-preceded carcinogenesis that result from immediate targeted DNA damage and persistent non-targeted effects. As described in Project 1, a single low dose of radiation administered during puberty increases mammary stem cell frequency in mice at adulthood. We have developed modeling approaches to dissect the mechanistic basis of this observation. Our approach used ABM that integrates experimental data obtained from in vivo expression profiles and in vitro growth analysis of human mammary epithelial cells exposed to specific conditions. In silico modeling of mammary morphogenesis using ABM allows these events to be evaluated in the context of development, with its incumbent complexity of lineage heterogeneity, proliferation and morphogenesis. It is not clear how processes induced by radiation affect stem cell regulation. Given the experimental challenges of testing whether radiation increases (1) cell inactivation (i.e. death) that then triggers proliferation; (2) signals for stem cell self-renewal, or (3) dedifferentiation, we turned to modeling to evaluate these potential mechanisms. Last year, we evaluated the contributions of radiation-induced cell death, senescence and self-renewal signaling. In the past year, we tested dedifferentiation and selective cell death as additional mechanisms for the increased mammary stem cell frequency in irradiated mice. We used these simulations to determine which mechanism when induced in a juvenile gland, had an impact on stem cell frequency measured at adulthood. The ABM rejected all modes of cell inactivation through cell death and replicative senescence as a possible means to increase stem cell frequency. Tumor incidence in P53 compromised mice exposed to different radiation qualities is now being modeled. P53 deficiency is simulated by reducing cell-cycle checkpoint of epithelial agents, leading to increase genomic instability, increased mitotic catastrophe, and reduced apoptosis. Consequently, dividing epithelial agents which are genomically unstable have a probability of mutating a set of functional targets (essential housekeeping functions, non-essential functions, and functions that dictate cellular phenotype). In silico tumors (clones growing out-of-control in the model) emerge from these simulations and can be tracked back to MaSC agents that have lost anti-growth signal sensitivity and extracellular matrix regulation via successive mutations. When genomic-instability is combined with the transient increase in stem-cell self-renewal following IR, our ABM predicts a shorter latency for tumor incidence, as observed experimentally.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
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